Deliverable D5.6

Report summarising the procedure for continuous stakeholder feedback gathering

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European Genomic Data Infrastructure

1. Executive Summary

This deliverable details the currently identified important technical stakeholders for GDI, as well as existing methods and processes for interacting with them, such as identifying key contacts between GDI and other important standards-setting organisations and stakeholders. Key technical developments in GDI are reported to the 1+MG stakeholder forum, including translation of technology advancements and collaboration opportunities from the external technical stakeholders,

The organisations/initiatives which we have identified as technical stakeholders are:

- Global organisations whose recommendations and standards GDI may implement:
    - **Global Alliance for Genomics and Health** (GA4GH); as the organisation establishing global standards for the management of genomic and health information
    - **Health Level Seven International** (HL7) for their HL7-FHIR™ standards, including resources and profiles, for exchanging information between clinical data resources.
    - The **International Organization for Standardization** (ISO) as organisation developing standards in genomics informatics
    - The **Observational Health Data Sciences and Informatics** network (OHDSI) for their standard and common data model supporting re-use of clinical information
    - The **Research Data Alliance** (RDA) as a way to develop and identify policies and solutions for data management and FAIR data
    - The **World Health Organisation** (WHO) who create specifications for genome data sharing
    - The **World Wide Web Consortium** (W3C) as the organisation developing standards and guidelines for the world wide web for interoperability, accessibility, privacy, internationalisation, and security
- European organisations / Laws / Initiatives for which GDI needs to consider how to interoperate:
    - The **Data Spaces Support Center** which facilitates the creation of sovereign, interoperable, and trustworthy multi-sector data spaces
    - The **European Commission**, as one of the drivers of pan-European Union requirements by setting standards for data spaces and identification of citizens.
    - The **European Health Data Space** (EHDS) legislation that establishes requirements for health-related data to be made accessible in the EHDS.
    - The **European Open Science Cloud** (EOSC), developing a platform for research data services in Europe.
    - **GAIA-X Health,** enabling citizen-centric, federated, sovereign health data spaces
    - The **GÉANT** Collaboration of European National Research and Education Networks (NRENs), for developing and maintaining interoperability standards in Information Technology.

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- European infrastructure projects related to GDI
  - **EOSC4Cancer**, a 'sister' project to GDI which brings in its use cases in GDI.
  - **EUCAIM,** developing a federated, secure, interoperable, privacy-preserving federated infrastructure for distributed analysis of cancer data
  - The **Genome of Europe** (GoE) project as one of the expected early data sources for data within the GDI
- Infrastructure services that GDI depends on:
  - The **Life Science Authentication and Authorisation infrastructure** (LSAAI) as a way to federate electronic identities across a federated life science infrastructure

We have established processes to keep in contact among these organisations and identified individuals within key standards-setting organisations and individuals within GDI to act as key contacts with many of them. We are continuing to work with WP4 (specifically T4.3 SOPs and T4.2 European Operations) to establish the processes and procedures to both receive and give feedback to relevant stakeholders.

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2. Contribution towards project outcomes

With this deliverable, the project has reached, or the deliverable has contributed to, the following project outcomes:

	Contributed
Outcome 1 Secure federated infrastructure and data governance needed to enable sustainable and secure cross border linkage of genomic data sets in compliance with the relevant and agreed legal, ethical, quality and interoperability requirements and standards based on the progress achieved by the 1+MG initiative.	Yes
Outcome 2 Platform performing distributed analysis of genetic/genomic data and any linked clinical/phenotypic information; it should be based on the principle of federated access to data sources, include a federated/multi party authorisation and authentication system, and enable application of appropriate secure multi-party and/or high-end computing, AI and simulation techniques and resources.	Yes
Outcome 3 Clear description of the roles and responsibilities related to personal data and privacy protection, for humans and computers, applicable during project lifetime and after its finalisation.	No
Outcome 4 Business model including an uptake strategy explaining the motivation, patient incentives and conditions for all stakeholders at the different levels (national, European, global) to support the GDI towards its sustainability, including data controllers, patients, citizens, data users, service providers (e.g., IT and biotech companies), healthcare systems and public authorities at large.	No
Outcome 5	No

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Sustained coordination mechanism for the GDI and for the GoE multi-country project launched in the context of the 1+MG initiative.	
Outcome 6 Communication strategy – to be designed and implemented at the European and national levels.	No
Outcome 7 Capacity building measures necessary to ensure the establishment, sustainable operation, and successful uptake of the infrastructure.	No
Outcome 8 Financial support to the relevant stakeholders to enable extension, upgrade, creation and/or physical connection of further data sources beyond the project consortium or to implement the communication strategy and for capacity-building.	No

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3. Methods

This deliverable is a report of the effort undertaken, as Task 5.2, to establish contacts with relevant technical stakeholder organisations. As part of WP5 it is deeply embedded in the technical pillar II, and the reporting and planning of activities of this task is taking place in the monthly Pillar II meetings.

Technical stakeholder-related activity is however distributed widely over the whole 1+MG initiative and the European Genomic Data Infrastructure (GDI) project. This ranges from legal interoperability with the European Health Data Space (EHDS), which is monitored from WP2 and the GDI Governance panel, via technical interoperability at the data and dataset level, which forms part of Pillar II (WP3 for genomics data standards; WP4 for data cataloguing standards; WP6 for data management), and technical interoperability at the experimental level in Pillar III (WP7 for the use-case specific metadata models and WP8 for the semantic embedding of those models in GDI). Technology advancement in Europe is scaled to a global audience via the Global Alliance for Genomics & Health (GA4GH) where GDI was recently selected as a driver project. Through this, GDI has a way to influence global technology standards required for secure data sharing. GDI technology implementations, thus have an impact on global interoperability of data services in genomics.

The use of standards within GDI is important to allow different nodes to use their own solutions of choice for supporting the required functionalities within GDI, as well as to ensure that the data within GDI are as FAIR as possible. Additionally it allows the infrastructure to evolve in a controlled and consistent way so that, as technologies and best practices change, GDI can adapt to support these as necessary, based on the 1+MG initiative data governance implementation driven from Pillar I, and prove the usefulness of technology by collaborating with the GDI use cases in Pillar III.

We have identified key contacts for different technical stakeholder organisations based on discussions within Pillar-wide and Task/WP-focused meetings in the project, especially within Pillar II which is tasked with technical implementation, deployment and operation of the GDI infrastructure. We must make use of relevant common standards when they are suitable to support the 1+MG mission and collaborate on changes and/or branch from the standard when we find them not suitable. If branching happens, it is beneficial to inform the global genomics community (GA4GH) about the gaps in the technology landscape that GDI has identified. Standards for the genomic data are very well established and only will need to be followed for new developments. Much more challenging are standards for associated clinical/phenotypic information which are essential for many types of use of the genomic information.

We monitor and maintain compatibility with relevant stakeholder organisations (when in alignment with 1+MG needs) by taking part in meetings and processes of those other organisations and

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reporting back in GDI. Relevant developments are reported back to regular Pillar, Work Package and Task meetings as required. Documentation of inputs is via the records of discussions in the various meeting agendas. Any relevant input to external organisations is fed back from these meetings after cascading to higher-level meetings within GDI to ensure consistent messaging.

4. Description of work accomplished

The work consisted of two main tasks: identifying the organisations relevant to interact with, and identifying one or more key individuals who take responsibility for the interactions. In addition, processes have been developed to incorporate feedback.

The key stakeholder organisations were classified into 4 groups: Global organisations whose recommendations and standards GDI may implement, European organisations / Laws / Initiatives for which GDI needs to consider how to interoperate, European infrastructure projects related to GDI, and Infrastructure services that GDI depends on.

4.1. Key Organisations

4.1.1. Global organisations whose recommendations and standards GDI may implement

4.1.1.1. GA4GH

GDI is now a GA4GH driver project. Several GDI participants are represented in various GA4GH working groups and report on their findings in monthly Pillar II meetings. GDI representatives have presented our progress at GA4GH meetings. Many of the GA4GH standards for data and metadata are implemented in the GDI.

GDI has been increasing the engagement with GA4GH as a driver project to bring the European use cases and requirements to a global audience. As a driver project GDI nominates two driver project champions, who are currently Melissa Konopko from the ELIXIR Hub and Dylan Spalding from CSC - IT centre for science Finland. This ensures that there is a clear and consistent way for GDI to interact with GA4GH at a high level. Additionally GDI is building a portfolio of experts in different standards, who can bring the depth of knowledge required to ensure that the GDI requirements are fully understood within GA4GH. To facilitate this a GA4GH Zenhub board[1] has been created, where issues relating to GA4GH standards and their deployment can be listed, prior to being reviewed and pruned before being brought to the relevant GA4GH workstreams. Other examples of evolving and developing standards where GDi can have an impact are the Federated Cohort Building (related to virtual cohorts within GDI) and the Experiments Metadata.

[1] https://app.zenhub.com/workspaces/ga4gh-driver-project-issues-651bdc8ea9094e05dd82801c/board

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European Genomic Data Infrastructure

Most important identified resources:

Resource	Link to information
Beacon	https://docs.genomebeacons.org/
VCF	https://samtools.github.io/hts-specs/VCFv4.3.pdf
CRAM	https://samtools.github.io/hts-specs/CRAMv3.pdf
BAM	https://samtools.github.io/hts-specs/SAMv1.pdf
Passport	https://github.com/ga4gh-duri/ga4gh-duri.github.io/blob/master/researcher_ids/ga4gh_passport_v1.md
AAI	https://ga4gh.github.io/data-security/aai-openid-connect-profile
Crypt4GH	https://doi.org/10.1093/bioinformatics/btab087
WES	https://github.com/ga4gh/workflow-execution-service-schemas
TES	https://ga4gh.github.io/task-execution-schemas/docs
refget	https://doi.org/10.1093/bioinformatics/btab524
htsget	https://samtools.github.io/hts-specs/htsget.html

4.1.1.2. HL7

The HL7 Fast Healthcare Interoperability Resources (FHIR) is a standards framework built from modular components. The FHIR Genomics standard defines FHIR representations for a range of genomic data structures (eg, variants, haplotypes, and variant implications), enabling a standards-based exchange of clinical data between systems.

Most important identified resources:

Resource	Link to information
FHIR Genomics	https://www.hl7.org/fhir/genomics.html
XPanDH project	https://xpandh-project.iscte-iul.pt/

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4.1.1.3. ISO

The International Standardization Organization (ISO) and the European Standardization Organization (CEN) provide standardisation documents on different phases of genomic data collection: sStandards for pre-analysis of samples, sStandards for library preparation and NGS analysis and. sStandards for NGS data . The most relevant to the goal of the GDI is ISO/TS 8376:2023 "Genomics informatics, Requirements for interoperable systems for genomic surveillance".

Most important identified resources:

Resource	Link to information
ISO/TC 215/SC 1: Genomics Informatics	https://www.iso.org/committee/7546903.html
ISO/TS 8376:2023 Genomics informatics, Requirements for interoperable systems for genomic surveillance	https://www.iso.org/standard/83159.html
ISO 4454: 2022, Genomics informatics, Phenopackets: A format for phenotypic data exchange	https://www.iso.org/standard/79991.html
ISO/TS 23357:2023 Genomics informatics — Clinical genomics data sharing specification for next-generation sequencing	https://www.iso.org/standard/75310.html

4.1.1.4. OHDSI

The Observational Health Data Sciences and Informatics network (OHDSI) maintains a standard and common data model (OMOP/CDM) supporting re-use of clinical information.

Most important identified resources:

Resource	Link to information
OMOP/CDM	https://www.ohdsi.org/data-standardization/

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4.1.1.5. RDA

The Research Data Alliance brings together policy makers, funders, publishers, infrastructure providers and researchers in a wide range of research fields to work on common and interoperable processes and standards. ELIXIR has a Focus Group which maintains structured contacts with RDA meetings and outputs.

Most important identified resources:

Resource	Link to information
Citing Dynamic Data	https://doi.org/10.15497/RDA00016
Persistent Identifiers	https://docs.pidinst.org/en/latest/ http://dx.doi.org/10.15497/RDA00027 https://doi.org/10.15497/rda00031

4.1.1.6. World Health Organisation

The World Health Organisation (WHO) creates specifications for genome data sharing.

Most important identified resources:

Resource	Link to information
Principles for human genome access, use and sharing	https://www.who.int/news-room/articles-detail/invitation-for-public-comment--who-principles-for-human-genome-access--use-and-sharing/

4.1.1.7. W3C

GDI is implementing the DCAT standard and OpenID Connect (OIDC) standards, in combination with the European application profile DCAT-AP to be compatible with the network of data cataloguing services that the EC is deploying. Our dataset-metadata efforts will be expressed as extensions to and compatible with DCAT AP and HealthDCAT-AP. The W3C OIDC standard is used for carrying electronic user authentication information between services, and the standard has been extended using the GA4GH Passport specification related to exchanging provenance and rights/permission information. These are also relevant to GDI cataloguing activities. The standard PROV-O is being considered for encoding of provenance metadata, and ODRL will be relevant for ELSI metadata. Other W3C standards may increase in relevance over the course of the project.

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Most important identified resources:

Resource	Link to information
DCAT	W3C latest published release: https://www.w3.org/TR/vocab-dcat-3/ EU DCAT-AP : https://joinup.ec.europa.eu/collection/semic-support-centre/solution/dcat-application-profile-data-portals-europe/releases
PROV-O	https://www.w3.org/TR/prov-o/
ODRL	https://www.w3.org/TR/odrl-model/
OIDC	https://openid.net/specs/openid-connect-core-1_0.html https://www.w3.org/wiki/WebID
Data authorisation	Part of OID4VC (https://openid.github.io/OpenID4VCI/openid-4-verifiable-credential-issuance-wg-draft.html) draft is not yet recognised in W3C - possibly through the GÈANT AARC TREE project. https://aarc-project.eu/ .

4.1.2. European organisations / Laws / Initiatives for which GDI needs to consider how to interoperate

4.1.2.1. Data Spaces Support Center

The Data Spaces Support Center facilitates the creation of sovereign, interoperable, and trustworthy multi-sector data spaces,

Most important identified resources:

Resource	Link to information
Home page	https://dssc.eu/

4.1.2.2. European Commission

The European Commission is an important driver of pan-European Union requirements for data spaces and data exchange, including standards for implementation and for identification and authorization of people who are allowed to access digital resources.

Most important identified resources:

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Resource	Link to information
DCAT-AP	https://ec.europa.eu/isa2/solutions/dcat-application-profile-data-portals-europe_en/
eDelivery	https://ec.europa.eu/digital-building-blocks/sites/display/DIGITAL/eDelivery
eIDAS	https://www.eid.as/
eWallet	https://ec.europa.eu/commission/presscorner/detail/en/ip_21_2663
SIMPL	https://digital-strategy.ec.europa.eu/en/policies/simpl

4.1.2.3. EHDS

Developments in the European Health Data Space (EHDS), both legally and technically, are followed by the project coordination as well as across all project Pillars according to their aims, and progress and interactions are reported in the Pillar, management board, and project meetings. 1+MG would have to fulfil connections to TEHDAS / Joint Action TEHDAS2 and Healthdata@EU projects that are preparing for the EHDS implementation; there are many people who are active in both 1+MG and one or more of these initiatives, and there is a genomics use case in Healthdata@EU.

1+MG Governance will be compatible with the EHDS and 1+MG aims to be established as an authorised participant in EHDS. This requires 1+MG data to be discoverable under EHDS, while access and processing can be provided using 1+MG Governance.

Most important identified resources:

Resource	Link to information
Regulation current status	https://www.consilium.europa.eu/media/70909/st07553-en24.pdf

4.1.2.4. European Open Science Cloud (EOSC)

EOSC is developing a platform for research data services in Europe. Liaison is organised through the many EOSC-related projects in which also members of the GDI team are participating.

4.1.2.5. GAIA-X Health

GAIA-X Health is moving towards citizen centric, federated, sovereign health data spaces under the technology umbrella of GAIA-X.

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Most important identified resources:

Resource	Link to information
Common data space for the health sector	https://gaia-x.eu/news-press/health-x-a-common-data-space-for-the-health-sector/

4.1.2.6. GÉANT

The collaboration of European National Research and Education Networks (NRENs), GÉANT, develops and maintains interoperability standards in Information Technology.
Especially relevant to GDI is the authentication and authorisation architecture blueprint project AARC, which we are connected to through our involvement in the LS-AAI.

Most important identified resources:

Resource	Link to information
AARC Community	https://wiki.geant.org/display/AARC

4.1.3. European infrastructure projects related to GDI

4.1.3.1. EOSC4Cancer

EOSC4Cancer is a European project running side by side to GDI which brings in its use cases in GDI. Contacts with this project are maintained by people who are involved in the use cases in both GDI and EOSC4Cancer. EOSC4Cancer is drafting a Roadmap, outlining key issues for a sustainable cancer data space beyond the project end – including access models, governance, data quality, security and privacy and others. Particularly, it will take into account developments in the EU Mission on Cancer, the European Health Data Space and related EU projects and research infrastructures.

4.1.3.2. EUCAIM

European Cancer Imaging Initiative (EUCAIM) is a project developing a federated, secure, interoperable, privacy-preserving federated infrastructure for distributed analysis of cancer imaging data.
GDI and EUCAIM has signed a Memorandum of Understanding (MoU), public announcement, pending, to recognise and strengthen the bilateral collaboration on key areas:

- **Knowledge sharing and exchanges:** Establishment and operation of multimodal data infrastructures for cancer, public and external affairs and other issues.
- **Strategic communication:** Joint strategic activities, e.g. joining stakeholders' events to increase the understanding and take on both infrastructure outputs.
- **Innovation solutions for tackling common challenges:** Explore technological solutions for the secure processing of sensitive data across the underlying

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infrastructure following the ongoing discussions of the European Health Data Space (EHDS) on Secure Processing Environments (SPEs). Moreover, the generation of multimodal synthetic data offers a great opportunity to explore the technical and governance interoperability of both infrastructures in preparation for the availability of real-world multimodal data.

- **Infrastructure services:** Explore and describe synergies between standards and open source references implementation that can be reused across both infrastructures, e.g. LS AAI, Beacon, Federated Analisis and Learning, driven by use cases that cut across both infrastructures and that provide evidence of the benefit for researchers, clinicians, innovators and policymakers.
- **Ethical, Legal, and Social Issues (ELSI):** discover and analyse ELSI aspects relevant for both infrastructures, working toward establishing interoperability governance when feasible.
- **Research data management:** Align on human genomics, biomolecular, cancer image data standards and other relevant international standards.
- **Training and capacity building:** Prepare joint training activities that will be pursued when needed.
- **Sustainability of the infrastructures:** Share and collaborate on long-term sustainability, especially in establishing one or more European Digital Infrastructure Consortium (EDIC).

Most important identified resources:

Resource	Link to information
Description of the initiative	https://digital-strategy.ec.europa.eu/en/policies/cancer-imaging

4.2.3.3. Genome of Europe

The Genome of Europe (GoE), through 1+MG WG12 is a relevant GDI use case, which is expected to be an important data source. A project proposal for the first phase of realising the Genome of Europe has been submitted, and a process for starting up the operation is in preparation. Already in this phase, the continued interaction of GDI and GoE is essential to make sure that mutual expectations between these projects are kept correct. Documenting the expectations will also help other use cases of the GDI/1+MG infrastructure in the future.

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4.2.4. Infrastructure services that GDI depends on

4.1.4.1. LifeScience AAI

The LifeScience Login[2], or Authentication and Authorisation Infrastructure (AAI), supports single sign on, allowing users to use their home institution credentials to log into the federated resources within GDI. It also is expected to help link to other identity and permissions management systems in other data spaces, such as EHDS via eID and eWallet, as well as other projects such as EOSC-Life[3] and infrastructures such as BBMRI-ERIC[4].

4.2. Summary of Key Contacts

For the technical stakeholders that we are most closely monitoring, one or more key liaisons have been identified and are given in the table below. For other technical stakeholders we identified, a responsible liaison is not considered essential and generic contacts are maintained. The best route to approach the liaisons is via the GDI coordination team (via gdi-coordination@elixir-europe.org), who will also be aware of the latest modifications and additions.

Stakeholder organisation	**GDI key contacts**
GA4GH	Melissa Konopko, Dylan Spalding to represent GDI as driver project; Giselle Kerry through ELIXIR:GA4GH Strategic partnership
HL7	<no individual liaison identified>
ISO	Ivo Gut (CNAG)
OHDSI	Jeroen Belien (HRI)
RDA	Gavin Farrell (ELIXIR) coordinates the RDA focus group. Bengt Persson (ICM) and Rob Hooft (HRI) have been directly involved in the past and can function as indirect liaisons.
World Health Organisation	Regina Becker (LNDS) and Tim Hubbard (ELIXIR)

[2] https://lifescience-ri.eu/ls-login/
[3] https://www.eosc-life.eu/
[4] https://www.bbmri-eric.eu/

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W3C	Rob Hooft (HRI)
Data Spaces Support Center	Juan Arenas (ELIXIR)
European Commission	Juan Arenas (ELIXIR) and Serena Scollen (ELIXIR)
EHDS	Regina Becker (LNDS) for data governance. Rob Hooft (HRI) for technical developments, Salvador Capella (BSC) for the 1+MG use case in Healthdata@EU
EOSC	Jonathan Tedds (ELIXIR)
GAIA-X Health	Juan Arenas (ELIXIR)
GÉANT	Dominik František Bučik (CZ)
EOSC4Cancer	Salvador Capella (BSC)
EUCAIM	Carles Hernandez-Ferrer (BSC), Juan Arenas (ELIXIR)
Genome of Europe	Jeroen van Rooij (Erasmus MC)
Life Science AAI	Tobias Petersen (DK) / Dominik František Bučik (CZ)

4.3. Managing feedback

Recording feedback and managing change processes is crucial in ensuring that stakeholder feedback is actioned in a timely and efficient way, as well as identifying issues and requirements that GDI has with the different standard setting organisations and for internal decision making processes, such as bringing the use case requirements from Pillar III to Pillars I and II.

4.3.1. Communications between the use cases and Pillars I and II

Pillar III includes WP7, which represents the use cases within GDI and brings their requirements to the technical infrastructure to make sure these requirements are met. An initial decision path[5] and associated form[6] has been designed to facilitate this communication process. In summary the decision path is:

1. Pillar III formulate a proposal internally, based on the use case requirements,
2. The form is completed and the proposal is discussed with Pillar II leads,

[5] https://docs.google.com/presentation/d/10BYx6vscDzPxs29QzS21XFEo-He8Qk1bXVBxEvBLpUU/edit
[6] https://docs.google.com/document/d/1PCYh_SyGbrccfaJJ_Bpa191cWUXvhysPNLVSOq1ajW8/copy

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3. WP3 review the proposal,
4. Product owners review the proposal,
5. If necessary, Pillar I, nodes, or other stakeholders review the proposal,
6. Pillar III is informed of the decision, and the reasoning.

It can be seen that the decision path specifically allows for relevant stakeholders and Pillars to be consulted on the proposal from the use cases (via Pillar III) or from Pillar III directly. The use of forms allows the decision to be recorded and annotated with any reasoning. Proposals can be recorded in GitHub / Zenhub and referred to in future if necessary.

4.3.2. Request For Comments

In conjunction with Task 4.3 "Standard Operating Procedures (SOPs)" and WP6 a proposal is being created for a formal Request For Comment[7] (RFC) process, based on those produced by the Internet Engineering Task Force (IETF), which allows extensive asynchronous discussion to be had over a technical issue, proposal, or standard. The RFCs have been used within GA4GH to create some of the technical standards, such as the Passport and AAI standards. They are designed for a specific audience and may not be suitable for all proposals and decisions, however they create a mechanism which records the process of making a decision, not only the outcome, which is important for a sustainable infrastructure.

RFCs help with change management within a federated infrastructure, and can also help when performing Privacy or Data Protection impact assessments (PIA / DPIA). Currently the proposal is that RFCs will only be used for breaking changes, new features, changing behaviour of an API or process, and the removal of features. They allow all stakeholders to be informed and have the opportunity to comment, supporting inclusivity.

The process is one of Proposal -> Iteration -> Acceptance or Rejection, and records the following in the outline:

- Name
- Summary
- Author(s)
- Motivation
- User Stories / Example
- Detailed Design
- Acceptance Criteria
- Unresolved Questions
- Prior Art
- Alternatives

[7] https://www.ietf.org/standards/rfcs/

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5. Results

While GDI is a European infrastructure, it should be interoperable with other infrastructures on a global scale to ensure maximum utility from the data can be achieved. Examples of this are the driver project status of GDI within GA4GH, and interest in GDI products and technologies expressed by countries outside Europe, such as Australia and Canada. It is also important to share our experience, and GDI can benefit from experience gained by other infrastructures and projects through organisations such as GA4GH. Utilising global standards means a European researcher should be able to use the same analysis pipeline or workflow on data both within the GDI and outside GDI, whether the data is in Europe or not.

As part of the interaction with GA4GH, GDI has presented at the GA4GH Connect[8] meeting in London in April 2023, the Plenary[9] meeting in San Francisco in September 2023, and will present requirements to a variety of work streams at the Connect meeting in April 2024. Additionally GDI, with its contacts within ELIXIR, can leverage the strategic partnership GA4GH and ELIXIR have, which in turn helps to ensure that the GDI is interoperable across ELIXIR and beyond, for example utilising the work done by the ELIXIR Tools, Compute, and Interoperability Platforms especially.

The ELIXIR Human Data Communities and Federated Human Data meetings have been leveraged to allow GDI updates to the wider community, and also for associated infrastructures and projects to present These meetings are open to all, and advertised in GDI communication channels. HDR-UK[10] has already presented, based on their TRE model and Australian Biocommons[11] have presented on their requirements: similarities were identified, both with products used (REMS[12]) and requirements, such as virtual cohorts and Beacon.

The GDI Starter Kit[13,14] is based on standards, with each product within the starter kit implementing one or more GA4GH standard, and it is this use of defined standards that allows heterogeneous implementation at different nodes while maintaining interoperability across the infrastructure, and with other infrastructures. It also helps lower the bar to implementation as it leverages existing standards and reference implementations to help nodes understand and feedback on the proposed technologies and standards.

Needs for LS AAI[15] have been identified (which is compatible with the GA4GH AAI and Passport standards), and there are collaborations to get things implemented that were lacking (e.g. for logging in to computers using LS AAI identifiers, to get guaranteed information about the level of assurance, and role based access control).

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8 https://docs.google.com/document/d/1WERL553lUduGlu-g9k92IB7Xy5Cg1q83jdyQekyyLFI/edit
9 https://www.ga4gh.org/document/meeting-report-11th-plenary/
10 https://www.hdruk.ac.uk/
11 https://www.biocommons.org.au/
12 https://github.com/GenomicDataInfrastructure/starter-kit-rems
13 https://github.com/GenomicDataInfrastructure/starter-kit
14 https://gdi.onemilliongenomes.eu/gdi-starter-kit.html
15 https://lifescience-ri.eu/ls-login/

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Through participation in standards-setting bodies we are able to influence the development of global standards, making sure that they incorporate our experience. An example of this is the development of the cluster of standards surrounding DCAT which predates the GDI project. Just after the publication of the FAIR Data Principles, the early DCAT standard that existed was seen as a good starting point for the development of a FAIR Data Point. Through participation in the DCAT development team for DCAT v2, the standard and the tool were made to fit together. Further adaptation by the EC and EHDS2-WP6 followed after that, and led to the adoption of the developing generic (EC: European DCAT-AP v3) and discipline-specific application profiles (EHDS2-WP6: Health DCAT-AP extension).

5. Discussion

The key organisations that are technical stakeholders in GDI have been identified, and liaison functions set up where necessary. Single contact persons have not been named for all stakeholders: many technical stakeholders beyond the ones originally envisioned at the time of the grant writing have been identified, all of which are relevant to the project, but not all of them warrant an individual liaison. On the other hand, for the most intense collaborative stakeholder relationships, multiple persons are named, and liaison tasks can be distributed to the individuals with the right specialisation. Where no people in the GDI project are directly in contact with a stakeholder, we have selected people who are in direct contact within their network with people who are.

Since we are operating in a moving landscape, we will closely need to follow any related developments (world wide, not only in nodes with advanced genome projects close to Europe like the UK, but also e.g. in Australia), and where needed add new relationships and liaisons in the future.

In conjunction with WPs 3, 4 and 6 in Pillar II change management processes need to be finalised to ensure that requirements from stakeholders are reviewed and implemented, and the results of the reviews recorded. Additional work on the RFC proposal, and the situations where it is required, needs to be completed. This will form part of the change management SOP that will be defined in T4.3.

Using global standards not only helps to ensure interoperability on a global scale, but it also helps to leverage existing work, expertise and experience in genomics and health from outside the European Union, where it conforms to the 1+MG ambition. This facilitates the deployment of the GDI within the proposed timescales, and helps Pillar II to demonstrate the capabilities and limitations of the infrastructure, hence identifying gaps and requirements which are tracked via Zenhub. The experience being gained as GDI is deployed to become an operational infrastructure can also be utilised to ensure that the standards used within GDI reflect the GDI and wider European requirements, such as alignment with the EHDS.

A formal proposal for GA4GH interactions is being worked on, to ensure that the relevant experts can both give and receive feedback on the GA4GH standards, identifying issues and requirements from a GDI perspective. Examples of the impact that GDI can have with respect to GA4GH standards development include GA4GH Federated Cohort Building, the GDI requirement for virtual cohorts to

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support the GDPR data minimisation principle, and requirements for interoperability between self-sovereign identities (eWallet) and GA4GH Passports.

6. Conclusions & Impact

Technical stakeholder relationships as described in this deliverable are coordinated through Pillar II monthly meetings and more specialised interactions discussed in individual work package meetings and task-level meetings throughout the GDI project.

Impact of the collaborations with stakeholders is expected to be two-sided: First, the GDI project will be benefiting from all the technical expertise that is involved in the technical stakeholders that have been identified. Keeping up to date in such a rapidly evolving field as genomic data sharing requires multiple contacts covering this complex area. Second, the experience gained through the GDI project in the building of the 1+MG infrastructure can be incorporated into European and world wide standards; especially in the European case also towards interdisciplinary infrastructure standards. Both of these directions will result in the optimal level of interoperability of the infrastructure built in GDI for the 1+MG initiative and the world around.

7. Next steps

Established contacts and internal reporting about these will be kept in place as established. We will continue to horizon-scan to identify any developments in the standards-setting environment and modify our strategy as applicable, as well as utilise the Human Data Communities communication channels and the ELIXIR::GA4GH strategic partnership.

During 2024 the SOPs will be finalised (D4.2) which will detail the change management processes and procedures to record, action, or finalise recommendations and feedback given and received by the different GDI stakeholders.

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